



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to TCS AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

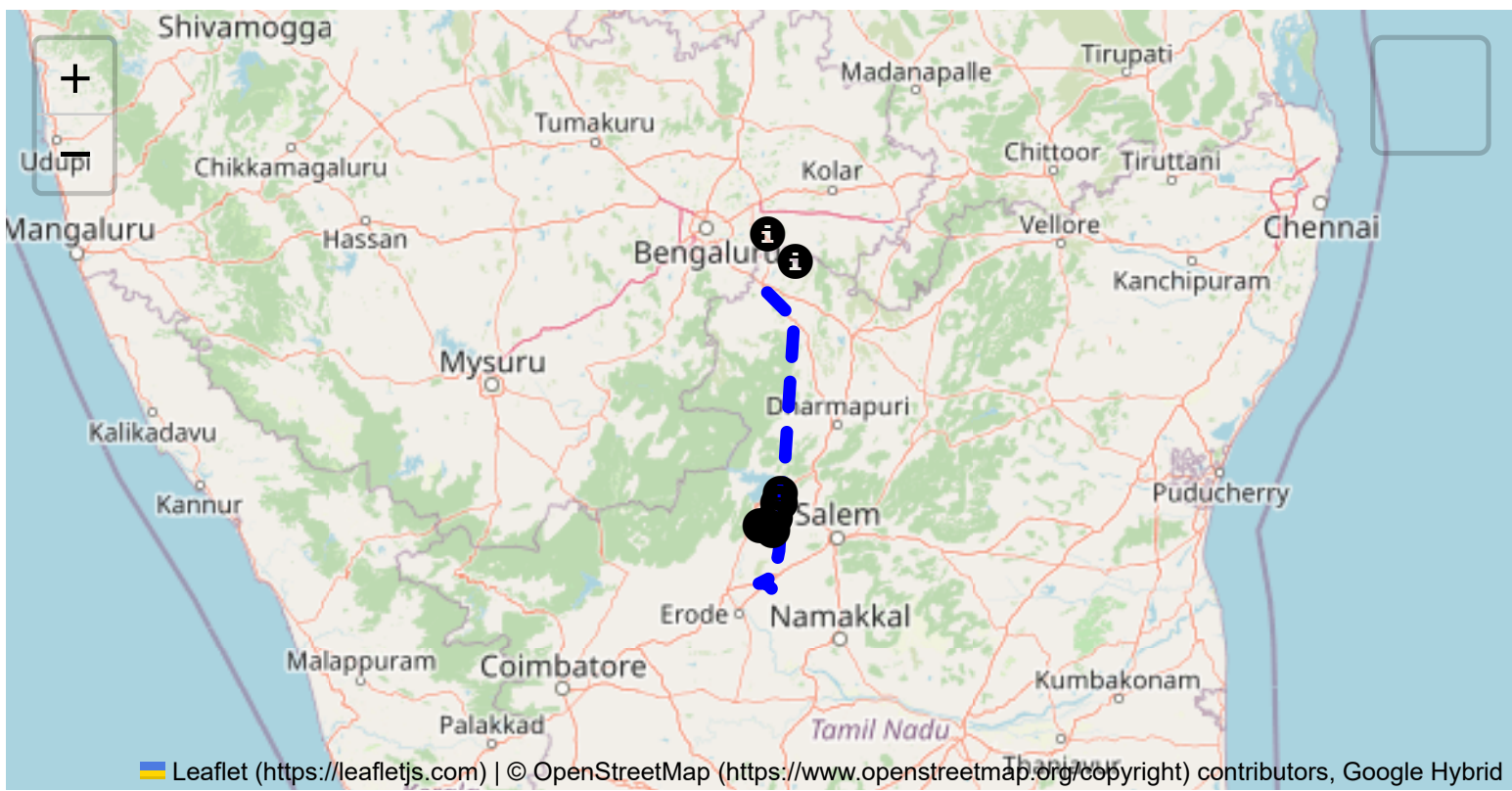
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

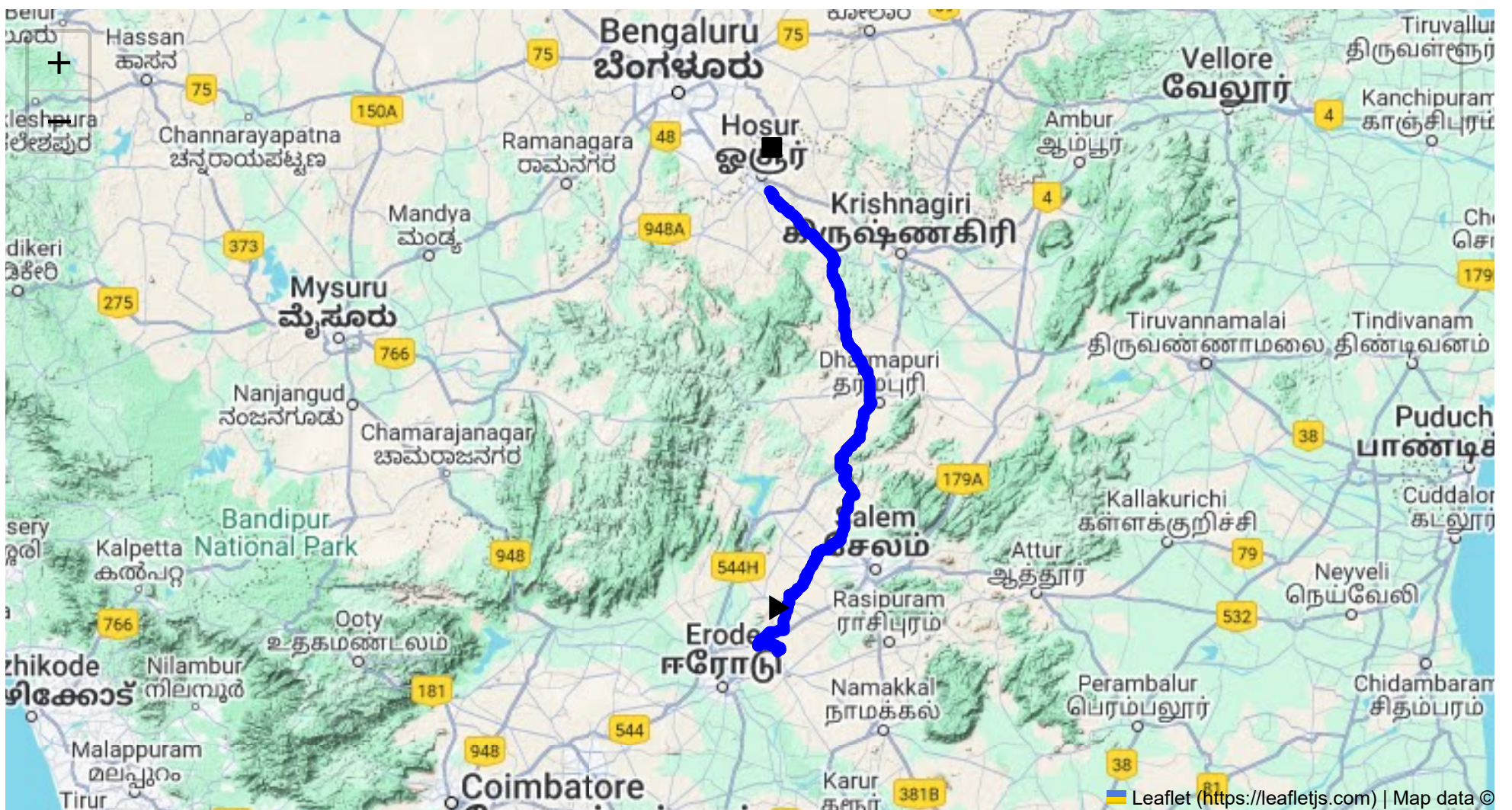
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 179.35 km
Estimated Duration: 3.6 hours
Adjusted Duration (Heavy Vehicle): 4.5 hours
Start: (11.4381, 77.8734)
End: (12.7011, 77.8476)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route starts at CVQF+23W, Sangagiri, and ends at PR2W+QP4, Karapalli, covering approximately 179.35 kilometers in Tamil Nadu, India. Key waypoints include sections along SH 86 and Bhavani Main Road, passing through towns like Sankari, Ayveli, and Erumaipatti. The route is characterized by local highways and well-maintained state roads.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Typical Weather:** The region experiences tropical weather with high temperatures and humidity, especially from March to June.
- **Monsoon Impact:** Heavy rains occur from June to September, increasing the risk of flooding and poor road visibility.
- **Fog and Dust:** Post-monsoon and winter months can see foggy conditions in the early mornings.

3. Analysis of Traffic Patterns

- **Peak Hours:** Traffic is generally heavy during morning (8-10 AM) and evening (5-7 PM) due to local commuter flow.
- **Congestion-Prone Areas:** Town centers in Sankari and near Bhavani Main Road intersections are frequently congested. Expect slow movement in these areas, particularly during peak times.

4. Assessment of Road Quality and Infrastructure

- **Road Surface:** The roads are predominantly paved, but some stretches may show signs of wear, particularly post-monsoon.
- **Infrastructure:** Adequate signage and road markings are generally present, but lighting may be poor in some rural sections.

5. Suggestions for Alternative Routes for Emergencies

- If the main route via SH 86 is obstructed, consider local roads connecting via minor state highways. However, these roads may be narrower and less maintained.
- Always verify these alternatives for clearance and suitability for heavy vehicles.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Permits:** Ensure all necessary permits for transporting hazardous materials are secured from the local authorities.
- **Curfews and Restrictions:** Some sections may impose restrictions on hazardous materials during peak hours or specific festivals.

7. Overview of Historical Incidents

- **Incidents:** Historical data indicates occasional accidents due to vehicle breakdowns and human error, but no significant hazardous materials incidents have been reported in recent years.
- **Preventive Measures:** Emphasize proper vehicle maintenance and driver training to mitigate risks.

8. Environmental Considerations and Sensitive Areas

- **Protected Areas:** Be aware of any nearby wildlife sanctuaries or protected areas, though the route primarily passes through developed regions.
- **Pollution Controls:** Follow local regulations regarding emissions and noise to reduce environmental impact.

9. Analysis of Communication Coverage

- **Coverage:** Mobile network coverage is generally reliable along the route, but rural areas may experience brief dead zones.
- **Recommendations:** Equip vehicles with offline navigation tools and maintain communication with base operations through radio as a backup.

10. Estimated Emergency Response Times

- **Urban Areas:** Response times are typically 15-30 minutes.
- **Rural Sections:** Could extend to 45-60 minutes depending on the distance from the nearest town or highway patrol.

12. Overall Summary of Risk Assessment

The route from Sangagiri to Karapalli is manageable with moderate risk levels. Key risks involve traffic congestion, especially around town centers, weather-related challenges during monsoon months, and occasional communication gaps in rural areas. Comprehensive preparation, including route planning and vehicle checks, can effectively mitigate potential hazards. Employing GPS tracking and maintaining emergency contacts is crucial for ensuring quick response to any incidents.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45352, 77.85698	15 KM/Hr
4	Turn	Medium	11.45764, 77.85219	30 KM/Hr
5	Turn	High	11.46303, 77.84945	15 KM/Hr
6	Turn	High	11.45571, 77.81702	15 KM/Hr
7	Turn	Medium	11.45654, 77.81706	30 KM/Hr
8	Turn	Medium	11.46916, 77.82431	30 KM/Hr
9	Blind Spot	Blind Spot	11.49395, 77.88537	10 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
10	Turn	High	11.49465, 77.88488	15 KM/Hr
11	Turn	Medium	11.56096, 77.89931	30 KM/Hr
12	Turn	Medium	11.57649, 77.90610	30 KM/Hr
13	Turn	Medium	11.59214, 77.90656	30 KM/Hr
14	Turn	Medium	11.63833, 77.94818	30 KM/Hr
15	Turn	High	11.64243, 77.95198	15 KM/Hr
16	Turn	Medium	11.64721, 77.95118	30 KM/Hr
17	Turn	Medium	11.67870, 77.96846	30 KM/Hr
18	Turn	High	11.70227, 77.97864	15 KM/Hr
19	Turn	Medium	11.71416, 77.99604	30 KM/Hr
20	Turn	Medium	11.71715, 77.99969	30 KM/Hr
21	Turn	Medium	11.71921, 78.01133	30 KM/Hr
22	Turn	Medium	11.72737, 78.02248	30 KM/Hr
23	Turn	High	11.73775, 78.03465	15 KM/Hr
24	Turn	High	11.73651, 78.03878	15 KM/Hr
25	Turn	High	11.74189, 78.04660	15 KM/Hr
26	Turn	Medium	11.77961, 78.05775	30 KM/Hr
27	Turn	Medium	11.86873, 78.08374	30 KM/Hr
28	Turn	Medium	11.87966, 78.08187	30 KM/Hr
29	Turn	Medium	11.92609, 78.06513	30 KM/Hr
30	Turn	Medium	11.93149, 78.06485	30 KM/Hr
31	Turn	Medium	11.93455, 78.06295	30 KM/Hr
32	Turn	Medium	11.93719, 78.05372	30 KM/Hr
33	Turn	Medium	11.95457, 78.04685	30 KM/Hr
34	Turn	Medium	11.95688, 78.04803	30 KM/Hr
35	Turn	High	11.95828, 78.04983	15 KM/Hr
36	Turn	Medium	11.95997, 78.04946	30 KM/Hr
37	Turn	Medium	11.96335, 78.05141	30 KM/Hr
38	Turn	High	11.96408, 78.05131	15 KM/Hr
39	Turn	Medium	11.96511, 78.04966	30 KM/Hr
40	Turn	High	11.96608, 78.04919	15 KM/Hr
41	Turn	Medium	11.96819, 78.05088	30 KM/Hr
42	Turn	Medium	12.12010, 78.13282	30 KM/Hr
43	Blind Spot	Blind Spot	12.51390, 78.03199	10 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
44	Turn	Medium	12.51369, 78.02985	30 KM/Hr
45	Turn	Medium	12.51644, 78.02566	30 KM/Hr
46	Turn	Medium	12.52955, 78.01978	30 KM/Hr
47	Turn	Medium	12.65389, 77.88732	30 KM/Hr
48	Turn	Medium	12.65815, 77.88249	30 KM/Hr
49	Turn	Medium	12.67918, 77.86202	30 KM/Hr
50	Turn	Medium	12.68940, 77.85544	30 KM/Hr
51	Turn	Medium	12.69420, 77.85429	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
3	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
4	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
5	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
9	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
10	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
12	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
13	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
14	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
15	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
16	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
17	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
18	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
19	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
20	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
21	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
23	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
25	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
26	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
27	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
28	hospital	Government Hospital, Jakkasamudram	12.3937246, 78.0522641	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
11	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
22	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
24	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
29	marketplace	Rayakottai tomato market	12.5237469, 78.0220377	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
30	school	Panchayat Union Primary School	12.6078768, 77.9381865	30 km/h	Medium
31	school	Panchayat Union Primary School	12.6835154, 77.8604798	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45764, 77.85219



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45571, 77.81702



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45654, 77.81706



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46916, 77.82431



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.49395, 77.88537



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49465, 77.88488



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56096, 77.89931



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.57649, 77.90610



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59214, 77.90656



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63833, 77.94818



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64243, 77.95198



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64721, 77.95118



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67870, 77.96846



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70227, 77.97864



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71416, 77.99604



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71715, 77.99969



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71921, 78.01133



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72737, 78.02248



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73775, 78.03465



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73651, 78.03878



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.74189, 78.04660



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.77961, 78.05775



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86873, 78.08374



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.87966, 78.08187



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.92609, 78.06513



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93149, 78.06485



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93455, 78.06295



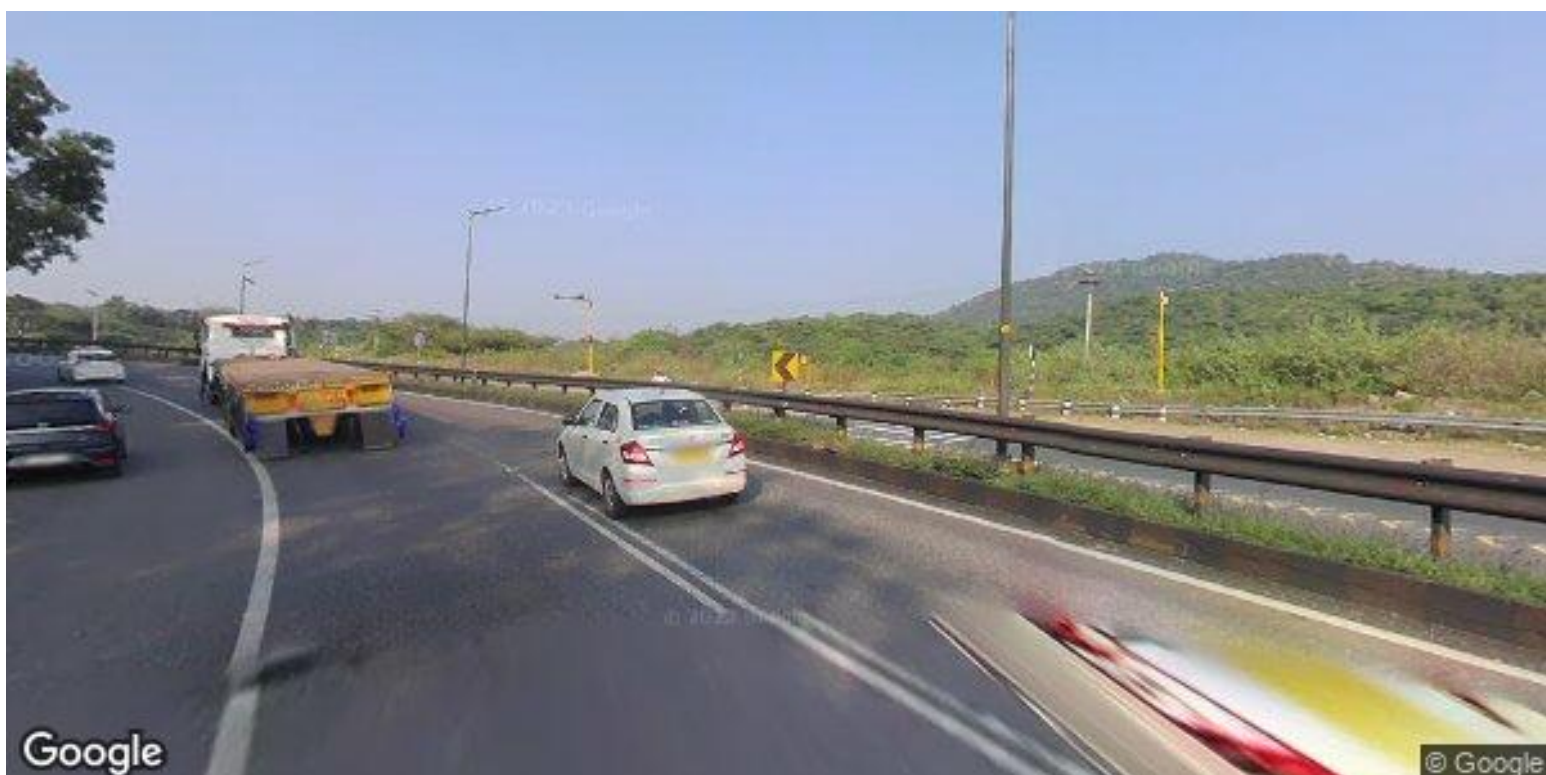
Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93719, 78.05372



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95457, 78.04685



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95688, 78.04803



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95828, 78.04983



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95997, 78.04946



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96335, 78.05141



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96408, 78.05131



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96511, 78.04966



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96608, 78.04919



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96819, 78.05088



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12010, 78.13282



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 12.51390, 78.03199



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.51369, 78.02985



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.51644, 78.02566



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.52955, 78.01978



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.65815, 77.88249



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.67918, 77.86202



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.68940, 77.85544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.69420, 77.85429
